

120 Days of Quantitative Finance

Week 16: Stochastic Calculus + Time Series

Day 108: Stochastic Volatility Models

1 Introduction

Traditional models such as Black-Scholes assume constant volatility, while GARCH models allow volatility to vary over time in discrete steps.

Stochastic volatility models take this further by modeling volatility itself as a random process in continuous time.

2 Motivation

Empirical observations in financial markets:

- volatility clustering
- leverage effect
- volatility smiles in option markets

These cannot be fully captured by constant volatility models.

3 General Framework

Asset price dynamics:

$$dS_t = \mu S_t dt + \sigma_t S_t dW_t^{(1)}$$

Volatility dynamics:

$$d\sigma_t = f(\sigma_t, t)dt + g(\sigma_t, t)dW_t^{(2)}$$

where:

- $W_t^{(1)}$ and $W_t^{(2)}$ may be correlated
- σ_t evolves stochastically

4 Heston Model

A widely used stochastic volatility model:

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^{(1)}$$

$$dv_t = \kappa(\theta - v_t)dt + \xi \sqrt{v_t} dW_t^{(2)}$$

$$\text{Corr}(dW_t^{(1)}, dW_t^{(2)}) = \rho$$

5 Interpretation of Parameters

- κ : speed of mean reversion
- θ : long-term variance
- ξ : volatility of volatility
- ρ : correlation between price and volatility

6 Key Features

- time-varying volatility
- mean-reverting variance
- correlation between returns and volatility (leverage effect)

7 Comparison with GARCH

- GARCH: discrete-time volatility modeling
- Stochastic volatility: continuous-time modeling

8 Financial Applications

- option pricing with volatility smiles
- risk management
- derivative modeling

9 Conclusion

Stochastic volatility models provide a more realistic representation of financial markets by treating volatility as a dynamic and random process.